Lab 1: Docker

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ITI _SYSTEM_ADMIN 46

ALEXANDRIA

Steps to install docker on RHEL 9:

1.

```
sudo dnf remove docker \
            docker-client \
            docker-client-latest \
            docker-common \
            docker-latest \
            docker-latest-logrotate \
            docker-logrotate \
            docker-engine \
            podman \
            runc
```

2.

```
sudo dnf -y install dnf-plugins-core
sudo dnf config-manager --add-repo
https://download.docker.com/linux/rhel/docker-ce.repo
```

3.

```
sudo dnf install docker-ce docker-ce-cli containerd.io
docker-buildx-plugin docker-compose-plugin
```

4.

```
sudo systemctl enable --now docker
sudo systemctl status docker
```

5.

```
sudo docker run hello-world
```

Pull nginx:alpine image:

$ sudo docker pull nginx:alpine
$ sudo docker run -d --name Ahmed-Kamel-ITI-46 nginx:alpine

Run Apache container with limit:

$ sudo docker run -d --name apache-limited --memory=70m --cpus=1 httpd

Get container ID:

$ sudo docker ps

Inspect container:

$sudo docker inspect <container_id>

```
Activities   Terminal                    Feb 26  21:49
root@serverA:/sys/fs/cgroup/system.slice/docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope
[root@serverA ~]# docker run -d --name Ahmed-Kamel-ITI-46 nginx:alpine
832c26dd4fa1ccaff4dab06010d5b8f24357f43ac7cca2e6db2c0ba44eb2236c
[root@serverA ~]# docker run -d --name apache-limited  --memory=70m --cpus=1 httpd
56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690
[root@serverA ~]# docker images
                                                                                    i Info →   U In Use
IMAGE               ID            DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  ef54e839ef54      21.8kB         9.52kB    U
httpd:latest        96b1e8f69ee3       174MB         47.6MB    U
nginx:alpine        1d13701a5f9f      91.9MB         26.9MB    U
nginx:latest        0236ee02dcbc       237MB         65.8MB    U
[root@serverA ~]# docker ps
CONTAINER ID   IMAGE         COMMAND                  CREATED         STATUS         PORTS     NAMES
56ac6916cd6e   httpd         "httpd-foreground"       34 seconds ago  Up 34 seconds  80/tcp    apache-limited
832c26dd4fa1   nginx:alpine  "/docker-entrypoint.…"   52 seconds ago  Up 52 seconds  80/tcp    Ahmed-Kamel-ITI-46
[root@serverA ~]# cd /sys/fs/cgroup/system.slice/docker
docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope/
docker-832c26dd4fa1ccaff4dab06010d5b8f24357f43ac7cca2e6db2c0ba44eb2236c.scope/
docker.service/
docker.socket/
[root@serverA ~]# cd /sys/fs/cgroup/system.slice/docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope/
[root@serverA docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope]# cat memory.max
73400320
[root@serverA docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope]# cat cpu.
cpu.idle        cpu.max         cpu.max.burst   cpu.stat        cpu.weight      cpu.weight.nice
[root@serverA docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope]# cat cpu.max
100000 100000
[root@serverA docker-56ac6916cd6e62b0bece3bbc1af3402d4da048a19dc0703705a2d81ec6773690.scope]#
```

Bonus part:

For being honest solved with the help of AI to find commands

```
[root@serverA ~]# docker run --name memory-eater --memory="50m" --memory-swap="50m" alpine sh -c "dd if=/dev/zero of=/dev/shm/fill_ram bs=1M count=100"
[root@serverA ~]# docker inspect memory-eater --format='OOMKilled: {{.State.OOMKilled}}, ExitCode: {{.State.ExitCode}}'
OOMKilled: false, ExitCode: 137
[root@serverA ~]# dmesg | grep -i "oom"
[ 5454.113399] dd invoked oom-killer: gfp_mask=0xcc0(GFP_KERNEL), order=0, oom_score_adj=0
[ 5454.113635]  oom_kill_process.cold+0xb/0x10
[ 5454.114396] [  pid  ]   uid  tgid total_vm      rss pgtables_bytes swapents oom_score_adj name
[ 5454.114402] oom-kill:constraint=CONSTRAINT_MEMCG,nodemask=(null),cpuset=docker-a7b7742bea14c521c381e3f34489b24b82d30a6d336849e7130926c9869e0326.scope,mems_
allowed=0,oom_memcg=/system.slice/docker-a7b7742bea14c521c381e3f34489b24b82d30a6d336849e7130926c9869e0326.scope,task_memcg=/system.slice/docker-a7b7742bea14c5
21c381e3f34489b24b82d30a6d336849e7130926c9869e0326.scope,task=dd,pid=8285,uid=0
[ 5454.114410] Memory cgroup out of memory: Killed process 8285 (dd) total-vm:2648kB, anon-rss:960kB, file-rss:880kB, shmem-rss:0kB, UID:0 pgtables:44kB oom_s
core_adj:0
[ 5880.343440] dd invoked oom-killer: gfp_mask=0xcc0(GFP_KERNEL), order=0, oom_score_adj=0
[ 5880.343470]  oom_kill_process.cold+0xb/0x10
[ 5880.343707] [  pid  ]   uid  tgid total_vm      rss pgtables_bytes swapents oom_score_adj name
[ 5880.343711] oom-kill:constraint=CONSTRAINT_MEMCG,nodemask=(null),cpuset=docker-f1128578a190c8273203f3fa8d585f2e6af2a10209c298f5c844a3d79a3ba419.scope,mems_
allowed=0,oom_memcg=/system.slice/docker-f1128578a190c8273203f3fa8d585f2e6af2a10209c298f5c844a3d79a3ba419.scope,task_memcg=/system.slice/docker-f1128578a190c8
273203f3fa8d585f2e6af2a10209c298f5c844a3d79a3ba419.scope,task=dd,pid=8728,uid=0
[ 5880.343718] Memory cgroup out of memory: Killed process 8728 (dd) total-vm:2648kB, anon-rss:964kB, file-rss:912kB, shmem-rss:0kB, UID:0 pgtables:44kB oom_s
core_adj:0
```